Question 1

Consider a long shell of radius R with a uniform surface charge density σ. Inside this cylindrical shell, coaxially placed, is a long solid cylinder of radius a ($a < R$) with a uniform volume charge density ρ.

(a) Find the electric field in the regions $0 \leqslant r \leq a$, $a \leqslant r \leq R$ and $r \geqslant R$ – 0.6 pts

(b) Determine the potential difference V between the surface of the solid cylinder (radius a) and the cylindrical shell (radius R) – 0.4 pts

Question 2

Consider a long, straight cylindrical conductor with radius R. The conductor carries a current I uniformly distributed across its cross-sectional area. Surrounding the conductor is a cylindrical shell with an inner radius a (where $a > R$) and outer radius b (where $b > a$). The shell carries a current $-I$ uniformly distributed across its cross-sectional area in the opposite direction.

(a) Find the magnetic field at a distance r from the axis of the conductor for the following regions: $0 \leqslant r \leq R$, $R \leqslant r \leq a$, $a \leqslant r \leq b$ and $r \geqslant b$ – 0.8 pts

(b) Sketch the magnetic field as a function of r – 0.2 pts

Question 3

Consider a circular loop of radius R made of a conducting wire, lying in the xy-plane with its center at the origin. The loop is placed in a time-varying magnetic field given by $\mathbf{B}(t) = B_0 \left(\frac{t}{\tau}\right) \hat{\mathbf{z}}$, where B_0 is a constant magnetic field strength, τ is a characteristic time, and t is time.

(a) Calculate the magnetic flux through the loop as a function of time – 0.2 pts

(b) Determine the induced electromotive force (emf) in the loop as a function of time – 0.2 pts

(c) Determine the direction of the induced current in the loop – 0.2 pts

(d) Calculate the induced current in the loop as a function of time given that the resistance of the loop is R – 0.2 pts

(e) If there is an additional constant magnetic field directed along the y-axis, $\mathbf{B}_c = B_y\hat{\mathbf{y}}$, what would be the magnitude and direction of the torque acting on the loop? – 0.2 pts

Question 4

Consider an electromagnetic wave propagating in vacuum with electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$. The wave is described by the following equations:

$$\mathbf{E}(z,t) = E_0 \cos(kz - \omega t)\hat{x}$$
$$\mathbf{B}(z,t) = B_0 \cos(kz - \omega t)\hat{y}$$

where E_0 and B_0 are the amplitudes of the electric and magnetic fields respectively, k is the wave number, and ω is the angular frequency. Here $c = \omega/k$, where c is the speed of light.

Show that these solutions satisfy Maxwell's equations in vacuum and/or show how E_0 and B_0 must be related in order to satisfy them.